

Beta regression – three views

1. Index form

The per-observation reading: what happens for each row i of your data.

$$\begin{aligned}y_i \mid \mu_i, \sigma_i &\sim \text{Beta}(\mu_i \sigma_i, (1 - \mu_i) \sigma_i) \\ \text{logit}(\mu_i) &= \beta_0 + \beta_1 x_i \\ \log(\sigma_i) &= \gamma_0\end{aligned}$$

2. Matrix form

The same model in matrix notation. The structural contract every textbook past chapter 4 switches to.

$$\begin{aligned}\mathbf{y} \mid \boldsymbol{\mu}, \boldsymbol{\sigma} &\sim \text{Beta}(\boldsymbol{\mu}, \boldsymbol{\sigma}) \\ \text{logit}(\boldsymbol{\mu}) &= \mathbf{X}\boldsymbol{\beta} \\ \log(\boldsymbol{\sigma}) &= \mathbf{Z}\boldsymbol{\gamma}\end{aligned}$$

3. Worked observation ($i = 1$)

The index-form equations evaluated at the first observation of your data, with coefficient estimates plugged in.

$$\underbrace{y_1}_{\text{observed}=0.175} = -0.824 + \underbrace{-0.0874 \cdot 1.43}_{\text{predicted}} + 1.13 = 0.175$$

Predicted $\hat{\mu}_1 = -0.95$; residual $\hat{\varepsilon}_1 = 1.13$.

And the σ submodel at $i = 1$:

$$\log \hat{\sigma}_1 = -1.04 \Rightarrow \hat{\sigma}_1 = \exp(-1.04) \approx 0.353$$